

Foundations of AI Algorithms (ST-1B)

1. Artificial Intelligence

AI is the ability of machines to **simulate human intelligence**.

Example systems include voice assistants and recommendation engines.

2. Characteristics of AI

- Learning
 - Reasoning
 - Problem solving
 - Perception
-

3. AI Problem Solving

AI agents search for solutions using **algorithms**.

State Space

Problem defined by:

- Initial state
 - Goal state
 - Operators
-

4. Search Algorithms

Breadth First Search (BFS)

Explores nodes **level by level**.

Advantages:

- Finds shortest path

Disadvantages:

- High memory usage
-

Depth First Search (DFS)

Explores **deep nodes first**.

Advantages:

- Uses less memory

Disadvantages:

- May get stuck in deep path